CLASS 8 — MATHS · FORMULA SHEET

Chapter 3: **A Story of Numbers** | Quick revision, one page

Roman numerals

- I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, M = 1000.
- I, X, C, M may be repeated up to 3 times; V, L, D are never repeated.
- Subtraction: I before V or X; X before L or C; C before D or M (e.g. IX = 9, XL = 40, CM = 900).
- Value = sum of symbol values, with a smaller symbol before a larger one subtracted.

Place value and bases

- In base b the digits are $0, 1, \ldots, b-1$ and the place values are $\ldots, b^3, b^2, b, 1$.
- $(d_3 d_2 d_1 d_0)_b = d_3 b^3 + d_2 b^2 + d_1 b + d_0$.
- Hindu-Arabic (decimal) system: base 10, uses the digits 0 to 9 and zero as a place holder.

Changing the base

- Decimal to base b: divide repeatedly by b; read the remainders from the last to the first.
- Base b to decimal: expand with powers of b and add.
- Largest k-digit number in base b is $b^k - 1$; there are b^k numbers with at most k digits.

Binary system

- Base 2, digits 0 and 1; place values $1, 2, 4, 8, 16, 32, \ldots$
- Binary addition: $1 + 1 = 10_2$, $1 + 1 + 1 = 11_2$ (carry 1).
- Computers store information in bits (binary digits).

Other ancient systems

- Babylonian system: base 60. Maya system: base 20 (dots, bars and a shell for zero).
- Brahmagupta (628 CE) gave rules for operations with zero and negative numbers.

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